

File Handling

Text Files

What is a File?

- When computer store data in hard disk, they are stored in the form of a file
- A computer file can be,
  - An audio file
  - A video file
  - A text file
  - Etc.
- Python allows accessing these files and manipulating them

File Access in Python

- Files can be text files or binary files
- Files can be opened with some options such as,
  - Read only
  - Write only
  - Append only
  - Read and/or write

Some Text File Open Modes

Modes	Details
r	Reading only. The file pointer at the BOF
r+	For reading and writing. The file pointer at the BOF
w	Writing only. If file exists, it'll be overwritten; If it doesn't a new file will be created for writing.
w+	For writing and reading. If file exists, it'll be overwritten; If it doesn't a new file will be created for reading and writing.
a	Appending. The file pointer is at EOF if the file exists. If the file does not exist, it creates a new file for writing.
a+	For appending and reading. The file pointer is at EOF if the file exists. If the file does not exist, it creates a new file for reading and writing.

Sample Program to Read a File

```
# Creating variables
fo = 0

# You have to have a file called
# MyText.txt with some text in the same folder

# Open a text file with
fo = open("MyText.txt", "r")
print ("File name : ", fo.name)
print ("Which mode? : ", fo.mode)
print ("What is in it?\n================")
print(fo.read())

# Accessing files in sub folder
fo2 = open("Sub\MyTextSub2.txt", "r")
print ("What is in it?\n================")
print(fo2.read())
```

You need some sample text files to test this!

Sample Program to Write to a File

```
# Creating variables
fo = 0

# You may have a file called
# MyText.txt with some text in the same folder

# Open a text file with write only mode
fo = open("MyText.txt", "w")
fo.write("Hello")
fo.close()
```

Reading a File - Line by Line

```
# Creating variables
fo = 0

# Open a text file with
fo = open("MyTextLines.txt", "r")
with fo:
    count = 1
    for line in fo:
        # Reading line by line
        print ("Line", count , "-", line, end="")
        count += 1
```

Reading a File - Character by Character

```
# Creating a variable
fo = 0

# Opening a text file
fo = open ("ReadThis.txt","r")

for x in range (39):
    print (fo.read(1), end="")
```

```
Contrary to popular belief
Lorem Ipsum
```

Seek a File - Part 1

```
# Creating variables
fo = 0

# Opening a text file read-binary
fo = open ("ReadThis.txt","r")
print ("My location : ", fo.tell())

# Read 10 characters only
print (fo.read(10))
print ("My location : ", fo.tell())

print (fo.read(3))
print ("My location : ", fo.tell())

# Seeking to 14th Location (absolute position)
fo.seek(14,0)
print ("My location : ", fo.tell())
print (fo.read(4))
print ("My location : ", fo.tell())
```

Seek a File - Part 2

```
fo.seek(21,0)
print ("My location : ", fo.tell())
print (fo.read(2))
print ("My location : ", fo.tell())

fo.seek(30,0)
print ("My location : ", fo.tell())
print (fo.read(4))
print ("My location : ", fo.tell())

fo.seek(0,2) # go to end of file
print ("My location : ", fo.tell())

fo.close()
```

How Seeking is Working

o	n	e		t	w	o		t	h	r	e	e		f	o	u	r	
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19 & 20

f	i	v	e		s	i	x		s	e	v	e	n		e	i	g	h	t	
21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41 & 42